Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.1 – Newton's First Law — Inertia in a Crash

Unit 2: Forces and Motion

Before you watch

You are about to watch part of a video filmed at a real crash-test laboratory, presented by a physics teacher called Dr. Griff Jones. Real cars are driven into a wall on purpose, so that engineers can study exactly what happens to the car and to the dummy inside it.

Listen for one word that comes up again and again: **inertia**.

While you watch

Fill these in as you go — you do not need full sentences yet.

1. What word does the video use for "the property of matter that resists a change in motion"?

   _____

2. A dummy is thrown forward in the demonstration with the tablecloth and the dishes. What does *that* demonstration show about objects that are **at rest**?

   _____

   _____

3. In the crash-test clip, the **car** stops suddenly when it hits the wall. Does the **dummy** stop at the same moment?

   _____

   _____

Newton's First Law

> **An object at rest will stay at rest, and an object in motion will stay in motion, unless acted upon by a net external force.**

Check: this law has two halves, and they are really the same idea. "Stays at rest" and "stays in motion" are both saying *nothing about the object's motion changes by itself* — something from outside has to push or pull on it first.

Balanced and unbalanced forces

Every object has forces acting on it all the time. What matters is whether those forces are **balanced** or **unbalanced**:

Forces are…	…so the object
balanced	keeps doing whatever it was already doing — staying still, or moving at a steady speed in a straight line.
unbalanced	changes its motion — it speeds up, slows down, or changes direction.

In the crash, the wall applies an unbalanced force to the car, so the car's motion changes very suddenly. The dummy only changes its motion once something — the seatbelt, or the steering wheel — applies an unbalanced force to *it* too.

Apply it

1) **State** Newton's First Law, in your own words.

> **State:** Give a specific name, value or other brief answer without explanation or calculation.

Model answer: An object at rest will stay at rest, and an object in motion will stay in motion, unless acted upon by a net external force.

2) **Describe** what happens to an unbelted passenger's body in the instant a car crashes into a wall.

> **Describe:** Give a detailed account or picture of a situation, event, pattern or process.

Model answer: The car stops very quickly because the wall applies a large force to it, but the passenger's body keeps moving forward at the car's original speed, because nothing has yet applied a force to the passenger — until they hit the dashboard, windscreen or steering wheel.

3) **Explain** why a passenger still needs a seatbelt even though the car has already come to a stop by the time the passenger is thrown forward.

> **Explain:** Give a detailed account including reasons or causes.

Model answer: By Newton's First Law, the passenger's body keeps its original motion until an unbalanced force acts on it directly. The car stopping does not by itself stop the passenger — their body only slows down once a force (the seatbelt, or something harder like the dashboard) actually acts on them.

Extension

Think further. The video also showed two eggs thrown at the same speed — one hits a hard wall, one hits a loose, hanging sheet. Both eggs end up stopped, but only one breaks. Using the idea of *how quickly* an unbalanced force acts, suggest why.

The wall stops the egg almost instantly, so the force needed is very large. The sheet gives way and stops the egg more gradually, over a longer time, so a much smaller force is enough — and the shell survives.